Research Methods and Stats for Psychologists

Dr. Margriet Groen

Overview

- Welcome!
- What this module is, how it runs, and how you'll be assessed

Fake news

- What is ‘true’ in psychological science?
- Parents and educators may believe ‘myths’ about child development without knowing whether they are supported by evidence.
- As scientists, it’s important to promote evidence-based practices and debunk false assumptions.



Myth or fact?

> “Showing cognitively stimulating videos to babies boosts their development?”

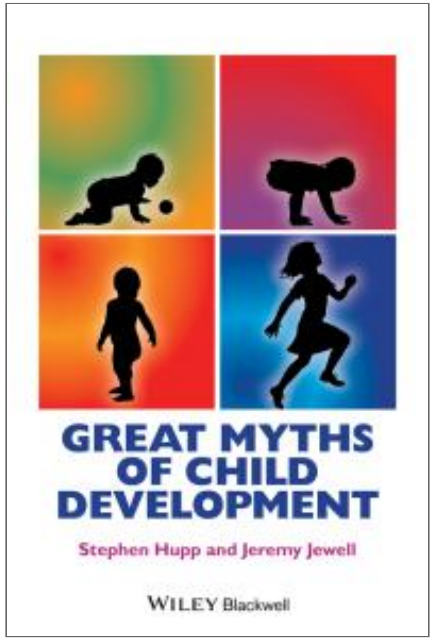
Myth!

- Watching a DVD 15 times over 6 weeks did not significantly increase word learning in 12- and 15-month-olds (Robb et al., 2009).
- 12–25-month-olds recognised a social partner and learned novel patterns via Face Time, but not pre-recorded videos (Myers et al., 2017).



Interested

If you are interested in developmental psychology and wish to learn more about myths that are debunked with more recent research:
Hupp, S., & Jewell, J. (2015). Great myths of child development (1st ed.). John Wiley & Sons.



What is this module about?

- This module introduces you to rigorous and transparent scientific practices.
- Using modern techniques, you will learn how to collect, present and analyse data.





How does it work?



Pre-recorded lectures



Readings



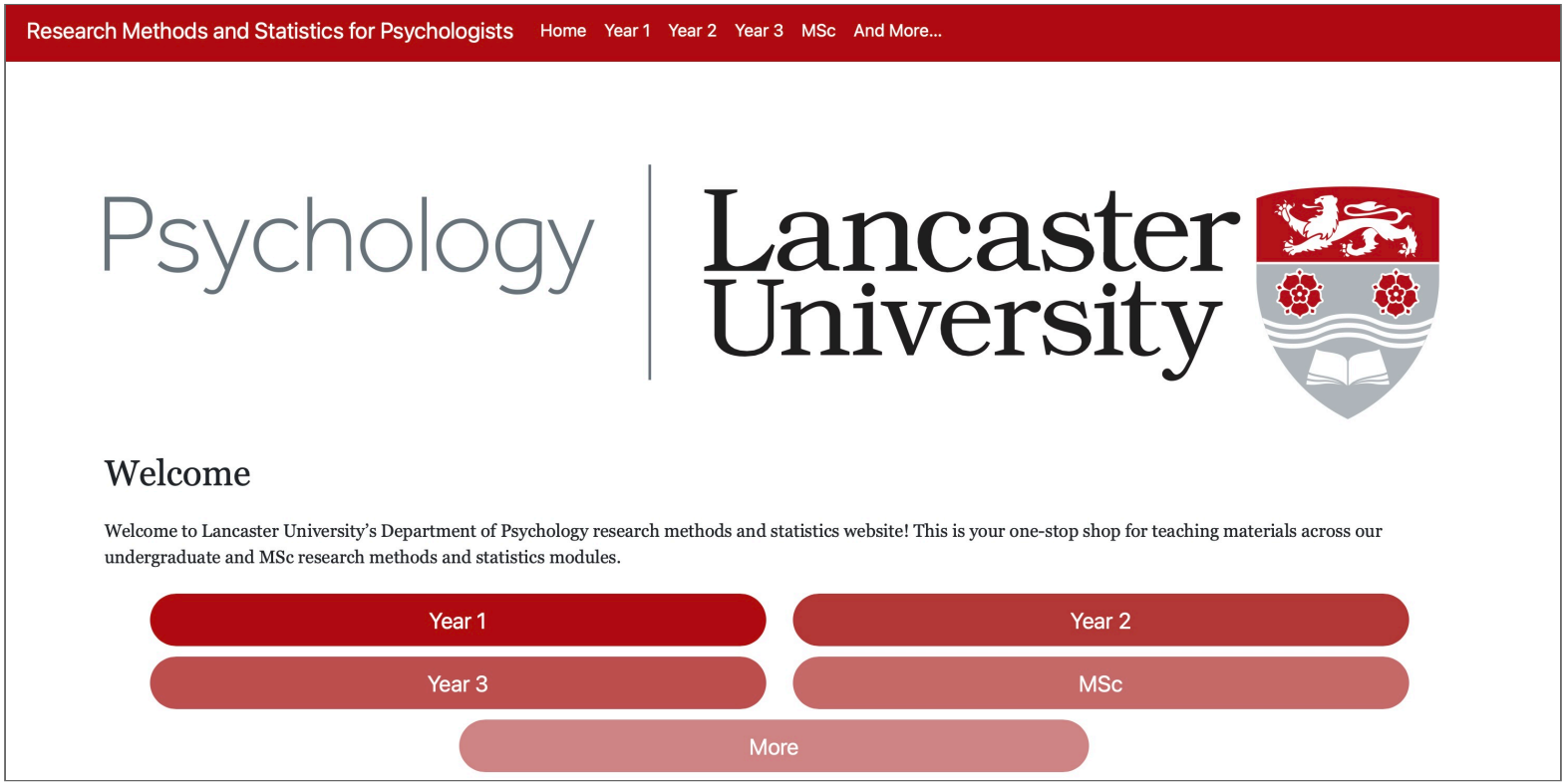
Practice tasks



Labs

What do you need?

- **The research methods and stats handbook https://lu-psy-r.github.io**
- The **R Studio server**: **https://ps-rstudio.lancaster.ac.uk/** You need to be on campus or using the **VPN** to login to the R Studio server.
- Library resources -> **Resource list**



Which topics will we cover?

- Describe the research proces, from concept design to post-publication
- Use specialist software to build relevant tools (e.g., questionnaires) for data collection.
- Report data in text, tables and visualisations, following relevant conventions and good practice.
- Identify statistical tests that are appropriate for a given design.
- Perform statistical analyses using a statistical software package, interpret the output and report the results following relevant conventions and good practice.
- Explain open science practices.
- Apply evidence-based reasoning and critically engage with empirical research.
- Participate in research both as an experimenter and a research study participant.

How does this module relate to the rest of the curriculum?



How will you be assessed?

Term	Module	Exam	Research report	Research participation
1	PSYC4141	50%	35%	15%
2	PSYC4242	50%	35%	15%
1	PSYC5141	60%	40%	
2	PSYC5242	60%	40%	
1 and 2	PSYC6041	dissertation		

How to ask for help

- **Ask your friends on your table** — working together during labs is encouraged
- **Ask a GTA or lecturer -** there are no stupid questions!
- **Ask on the Q & A Forum on Moodle -** if you are wondering other are probably too
- **Ask during the live online Q & A session -** link is on Moodle